Software Design (Lecture 4)

Prof. H S Behera

Organization of this Lecture

- Brief review of previous lectures
- Introduction to Software design
- Goodness of a design
- Functional Independence
- Cohesion and Coupling
- Function-oriented design vs. Object-oriented design
- Summary

Review of previous Lectures

- Introduction to software engineering
- Life cycle models
- Requirements Analysis and Specification:
  - ✓ Requirements gathering and analysis
  - ✓ Requirements specification

Introduction to SD

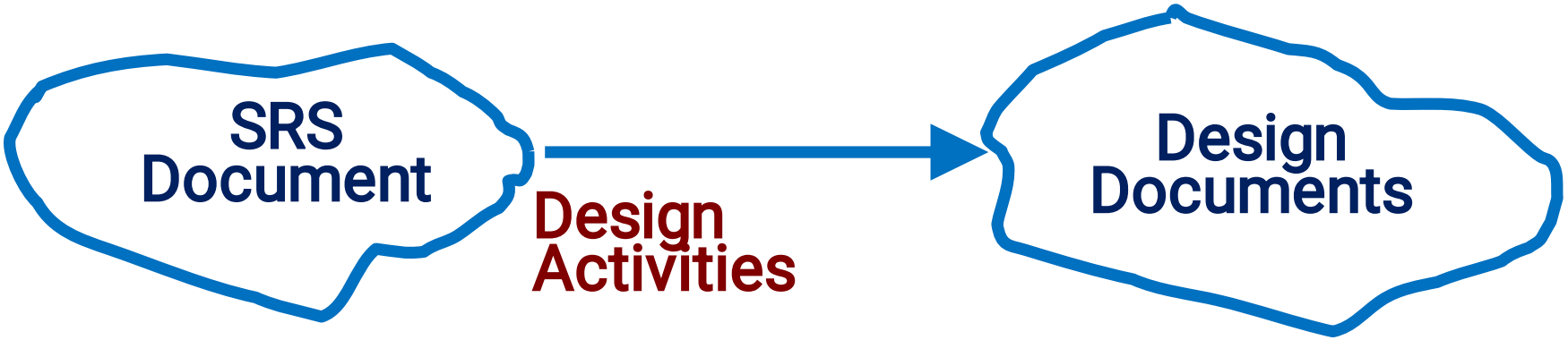
- **Software design** is the phase where technical requirements (**SRS**) are transformed into a blueprint for constructing the software, **focusing on system architecture, components, interfaces, and data flow.**
- It acts as a bridge between problem understanding and code implementation, aiming to create scalable, maintainable, and efficient systems while reducing complexity.

Introduction to SD

- **software design (SD) transforms requirement specifications (SRS) into an implementable blueprint, focusing on module structure, interfaces, data structures, and algorithms.**
- Key principles include achieving high cohesion and low coupling to improve maintainability, alongside user-centric design and systematic process modeling.

Introduction

- Design phase transforms SRS document:
  - into a form easily implementable in some programming language.



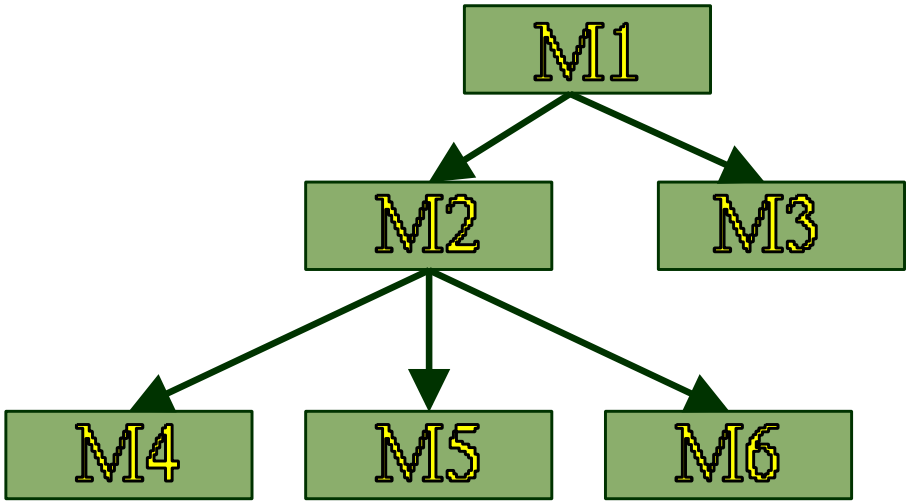
Items Designed During Design Phase

- **Module structure,**
- **Control relationship among the modules**
  - call relationship or invocation relationship
- **Interface among different modules,**
  - data items exchanged among different modules,
- **Data structures of individual modules**
- **Algorithms for individual modules.**

Module structure

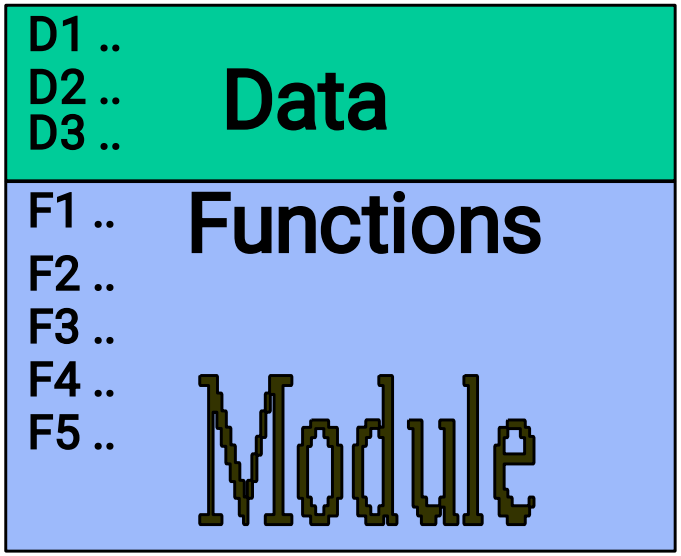
- **Module structure in software design, involves decomposing a complex system into independent, manageable units (modules) to enhance maintainability, scalability, and reusability**.
- It breaks a system into smaller, cohesive units with low coupling, often depicted using hierarchical structure charts that define control relationships and interfaces.

Module Structure



Introduction

- **A module consists of:**
  - several functions
  - associated data structures.



Introduction

- **Good software designs:**
  - seldom arrived through a single step procedure:
  - but through a series of steps and iterations.

Introduction

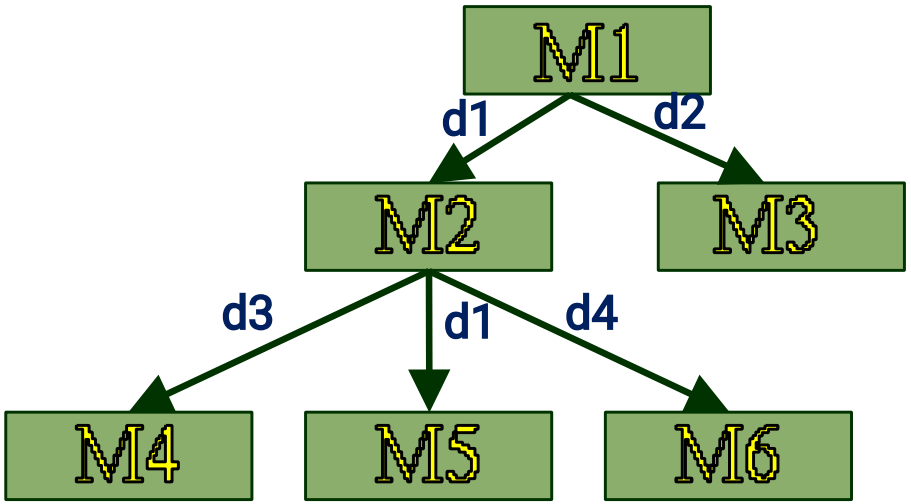
- **Design activities are usually classified into two stages:**
  - **preliminary (or high-level) design**
  - **detailed design.**
- Meaning and scope of the two stages:
  - vary considerably from one methodology to another.

High-level design

- **High-level design** or **HLD** is an initial step in the development of applications where the overall structure of a system is planned which mainly focuses on how different components of the system work together without getting to know about internal coding and implementation.
  - Helps everyone involved in the project to understand the goals and ensures good communication during development.
  - Crucial for developers, architects, and product managers because it allows them to make sure that all stakeholders are aligned with the project objectives. That's why it is also known as **macro-level design.**

High-level design

- **Identify:**
  - ✓Modules
  - ✓Control relationships among modules
  - ✓Interfaces among modules.



High-level design

- **The outcome of high-level design (HLD) is the system architecture or program structure**, usually documented as a blueprint.
- It decomposes the system into high-level, cohesive components and defines their interactions, data flows, and interfaces, bridging the gap between requirements and detailed design.

High-level design

- **The outcome of high-level design:**
  - **program structure (or software architecture).**
  - Software architecture (often referred to as program structure) is the **fundamental organization** of a software system.

High-level Design

- **Several notations are available to represent high-level design**:
  - Usually a tree-like diagram called **structure chart** is used.
  - **Other notations**:
    - Jackson diagram or Warnier-Orr diagram can also be used.

Detailed design

- **Detailed design** in software engineering is the low-level design phase (often called **Low-Level Design or LLD**) that transforms high-level architectural blueprints into detailed, actionable specifications for coding.
- It defines the internal logic, data structures, algorithms, interface details, and class hierarchies of each module, ensuring a precise blueprint for developers to write, test, and implement code.

Detailed design

- For each module, design:
  - data structure
  - algorithms
- Outcome of detailed design:
  - ✓module specification.

A fundamental question:

- **How to distinguish between good and bad designs?**
  - Unless we know what a good software design is:
    - ✓ we can not possibly design one.

Good and bad designs

- There is no unique way to design a system.
- Even using the same design methodology:
  - ✓ different engineers can arrive at very different design solutions.
- We need to distinguish between **good and bad designs**.

What Is Good Software Design?

- Should implement all functionalities of the system correctly.
- Should be easily understandable.
- Should be efficient.
- Should be easily amenable to change,
  - i.e. easily maintainable.

What Is Good Software Design?

- Understandability of a design is a major issue:
  - determines goodness of design:
  - a design that is easy to understand:
    - ✓also easy to maintain and change.

What Is Good Software Design?

- Unless a design is easy to understand,
  - tremendous effort needed to maintain it
  - We already know that about 60% effort is spent in maintenance.
- If the software is not easy to understand:
  - maintenance effort would increase many times.

Understandability

- Use consistent and meaningful names
  - for various design components,
- Design solution should consist of:
  - a cleanly decomposed set of modules (**modularity**),
- Different modules should be neatly arranged in a hierarchy:
  - in a neat tree-like diagram.

Modularity

- **Modularity** is the design principle of breaking a complex system into smaller, independent, and manageable components (modules) that interact through well-defined interfaces.
- By promoting **high cohesion and low coupling**, modularity increases reusability, simplifies maintenance, enhances scalability, and facilitates parallel development

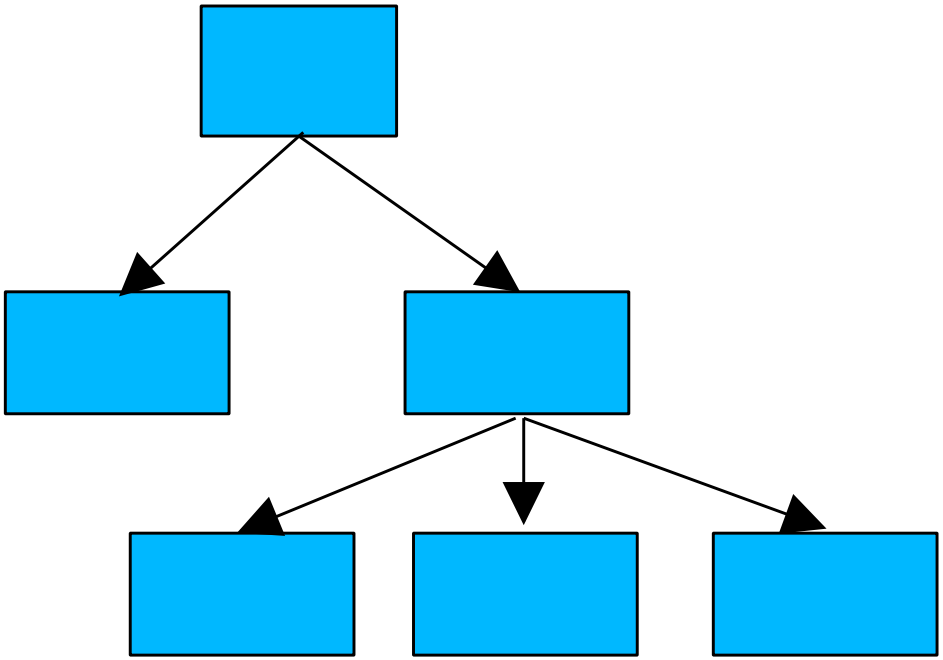
Modularity

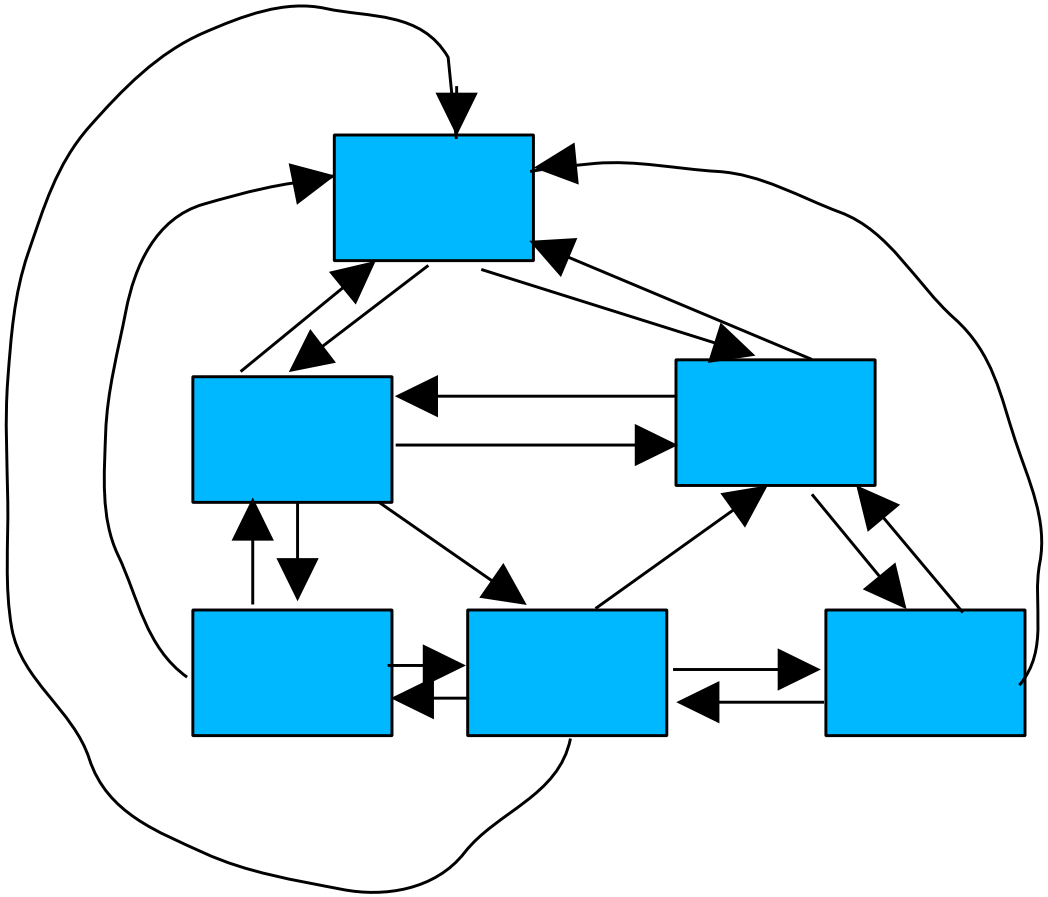
- **Modularity is a fundamental attributes of any good design.**
  - ✓ Decomposition of a problem cleanly into modules:
  - ✓ Modules are almost independent of each other
  - ✓ divide and conquer principle.

Modularity

- **If modules are independent:**
  - modules can be understood separately,
    - ✓reduces the complexity greatly.
  - To understand why this is so,
    - ✓remember that it is very difficult to break a bunch of sticks but very easy to break the sticks individually.

Example of Cleanly and Non-cleanly Decomposed Modules





Modularity

- **In technical terms, modules should display:**
  - **high cohesion**
  - **low coupling.**
- We will shortly discuss:
  - ✓cohesion and coupling.

Modularity

- Neat arrangement of modules in a hierarchy means:
  - **low fan-out**
  - **abstraction**

Low fan-out

- **low fan-out** refers to a design where a specific module, class, or function depends on or interacts with only a small number of other modules. It is generally considered a hallmark of good software architecture because it indicates high cohesion and low coupling.
- It is a key metric used to evaluate software complexity and maintainability

Abstraction

- **Abstraction** is the practice of simplifying complex systems by focusing on essential features while hiding unnecessary, underlying implementation details. It acts as a powerful tool for managing complexity, fostering code reusability, and ensuring maintainable software architecture.
- **Abstraction** is a fundamental principle used to manage complexity by focusing on the " what" of a system while hiding the "how" of its implementation.

Cohesion and Coupling

- **Cohesion and coupling** are fundamental software design principles used to measure code quality. **High cohesion** means a module focuses on a single task, enhancing maintainability. **Low coupling** means modules are independent, reducing dependency risks.
- The goal is to achieve high cohesion and low coupling to build manageable, scalable, and resilient/flexible systems.

Cohesion and Coupling

- Cohesion is a measure of:
  - functional strength of a module.
  - A cohesive module performs a single task or function.
- Coupling between two modules:
  - a measure of the degree of interdependence or interaction between the two modules.

Cohesion and Coupling

- **A module having high cohesion and low coupling:**
  - **functionally independent of other modules:**
    - ✓ A functionally independent module has minimal interaction with other modules.

Advantages of Functional Independence

- **Better understandability and good design:**
- **Complexity of design is reduced,**
- **Different modules easily understood in isolation:**
  - ✓modules are independent

Advantages of Functional Independence

- Functional independence reduces error propagation.
  - degree of interaction between modules is low.
  - an error existing in one module does not directly affect other modules.
- Reuse of modules is possible.

Advantages of Functional Independence

- A functionally independent module:
  - can be easily taken out and reused in a different program.
    - ✓each module does some well-defined and precise function
    - ✓the interfaces of a module with other modules is simple and minimal.

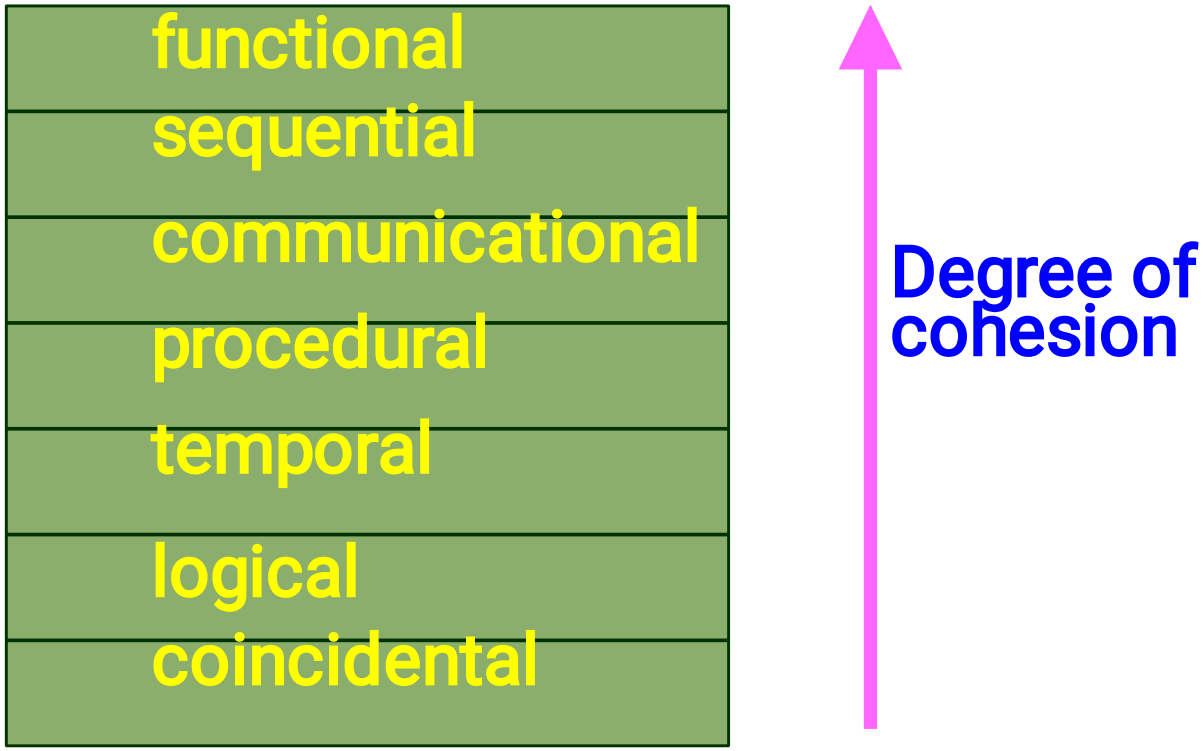
Functional Independence

- **Unfortunately, there are no ways:**
  - to quantitatively measure the degree of cohesion and coupling:
  - classification of different kinds of cohesion and coupling:
    - will give us some idea regarding the degree of cohesiveness of a module.

Classification of Cohesion

- **Classification is often subjective:**
  - yet gives us some idea about cohesiveness of a module.
- By examining the type of cohesion exhibited by a module:
  - we can roughly tell whether it displays high cohesion or low cohesion.

Classification of Cohesiveness



Coincidental cohesion

- The module performs a set of tasks:
  - which relate to each other very loosely, if at all.
    - the module contains a random collection of functions.
    - functions have been put in the module out of pure coincidence without any thought or design.

Logical cohesion

- All elements of the module perform similar operations:
  - e.g. error handling, data input, data output, etc.
- An example of logical cohesion:
  - a set of print functions to generate an output report arranged into a single module.

Temporal cohesion

- **The module contains tasks that are related by the fact:**
  - ✓all the tasks must be executed in the same time span.
- Example:
  - The set of functions responsible for
    - ✓initialization,
    - ✓start-up, shut-down of some process, etc.

Procedural cohesion

- The set of functions of the module:
  - all part of a procedure (algorithm)
  - certain sequence of steps have to be carried out in a certain order for achieving an objective,
    - e.g. the algorithm for decoding a message.

Procedural cohesion

- **Procedural cohesion** occurs when elements within a software module are grouped together simply because they must be executed in a specific sequential order to complete a process, rather than sharing a single functional purpose. It is characterized by following a procedural flow, such as a sequence of steps or a loop.

Communicational cohesion

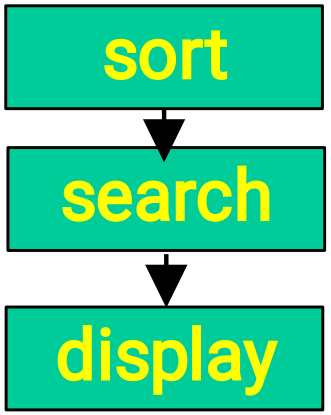
- All functions of the module:
  - reference or update the same data structure,
- Example:
  - the set of functions defined on an array or a stack.

Communicational cohesion

- **Communicational cohesion** occurs when modules are grouped together because they operate on the same input/output data, such as methods managing a single record or file. It is a strong form of cohesion that improves readability and maintainability, ranking high in the hierarchy below functional and sequential cohesion.

Sequential cohesion

- Elements of a module form different parts of a sequence,
  - output from one element of the sequence is input to the next.
  - Example:



Functional cohesion

- Different elements of a module cooperate:
  - to achieve a single function,
  - e.g. managing an employee's pay-roll.
- When a module displays functional cohesion,
  - we can describe the function using a single sentence.

Determining Cohesiveness

- Write down a sentence to describe the function of the module
  - If the sentence is compound,
    - ✓it has a sequential or communicational cohesion.
  - If it has words like “first”, “next”, “after”, “then”, etc.
    - ✓it has sequential or temporal cohesion.
  - If it has words like initialize,
    - ✓it probably has temporal cohesion.

Coupling

- Coupling indicates:
  - how closely two modules interact or how interdependent they are.
  - The degree of coupling between two modules depends on their interface complexity.

Coupling

- There are no ways to precisely determine coupling between two modules:
  - ✓classification of different types of coupling will help us to approximately estimate the degree of coupling between two modules.
- Five types of coupling can exist between any two modules.

Classes of coupling

Data coupling

- Two modules are data coupled,
  - if they communicate via a parameter:
    - ✓ an elementary data item,
    - ✓ e.g an integer, a float, a character, etc.
  - The data item should be problem related:
    - ✓ not used for control purpose.

Stamp coupling

- Two modules are <u>stamp coupled</u>,
  - if they communicate via a composite data item
    - ✓such as a record in PASCAL
    - ✓or a structure in C.

Control coupling

- Data from one module is used to direct
  - order of instruction execution in another.
- Example of control coupling:
  - a flag set in one module and tested in another module.

Common Coupling

- Two modules are <u>common coupled</u>,
  - if they share some global data.

Content coupling

- Content coupling exists between two modules:
  - if they share code,
  - e.g, branching from one module into another module.
- The degree of coupling increases
  - from data coupling to content coupling.

Neat Hierarchy

- Control hierarchy represents:
  - organization of modules.
  - control hierarchy is also called program structure.
- Most common notation:
  - a tree-like diagram called structure chart.

Neat Arrangement of modules

- Essentially means:
  - low fan-out
  - abstraction

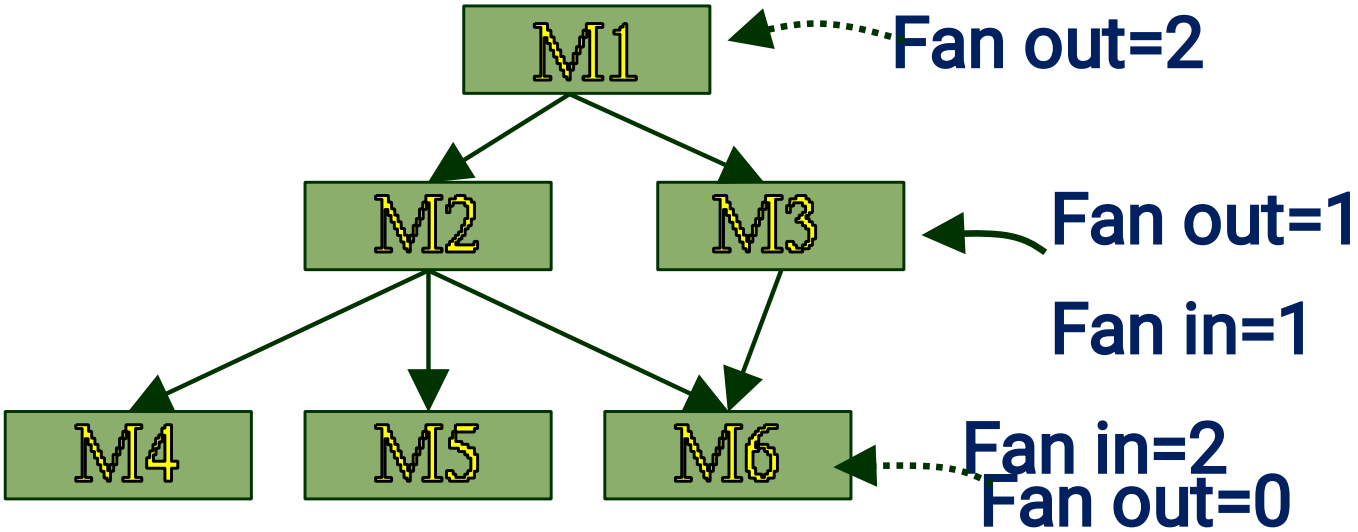
Characteristics of Module Structure

- Depth:
  - number of levels of control
- Width:
  - overall span of control.
- Fan-out:
  - a measure of the number of modules directly controlled by given module.

Characteristics of Module Structure

- Fan-in:
  - indicates how many modules directly invoke a given module.
  - High fan-in represents code reuse and is in general encouraged.

Module Structure



Goodness of Design

- A design having modules:
  - with high fan-out numbers is not a good design:
  - a module having high fan-out lacks cohesion.

Goodness of Design

- A module that invokes a large number of other modules:
  - likely to implement several different functions:
  - not likely to perform a single cohesive function.

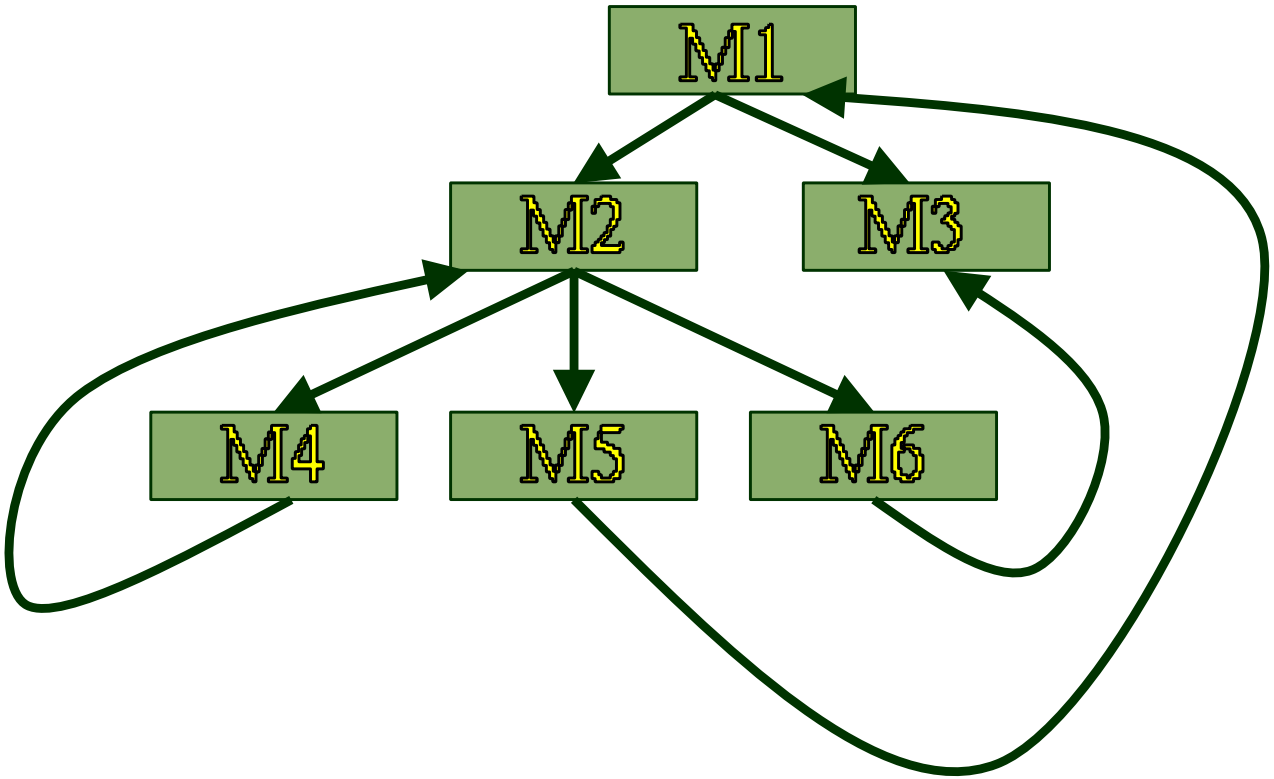
Control Relationships

- A module that controls another module:
  - said to be superordinate to it.
- Conversely, a module controlled by another module:
  - said to be subordinate to it.

Visibility and Layering

- A module A is said to be visible by another module B,
  - if A directly or indirectly calls B.
- The layering principle requires
  - modules at a layer can call only the modules immediately below it.

Bad Design



Abstraction

- Lower-level modules:
  - do input/output and other low-level functions.
- Upper-level modules:
  - do more managerial functions.

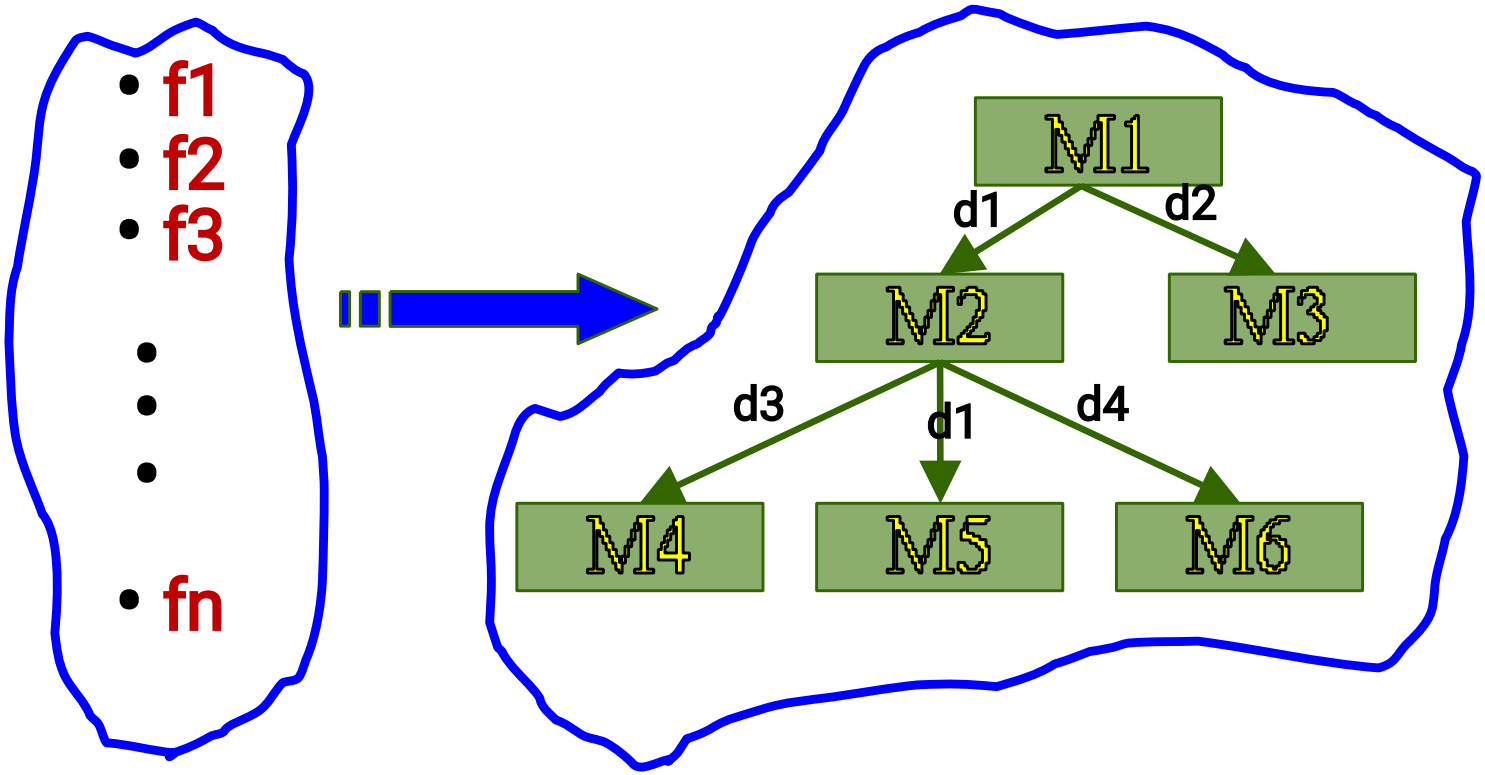
Abstraction

- The principle of abstraction requires:
  - ✓lower-level modules do not invoke functions of higher level modules.
  - ✓Also known as layered design.

High-level Design

- High-level design maps functions into modules {fi} {mj} such that:
  - Each module has high cohesion
  - Coupling among modules is as low as possible
  - Modules are organized in a neat hierarchy

High-level Design



Design Approaches

- Two fundamentally different software design approaches:
  - Function-oriented design
  - Object-oriented design

Design Approaches

- These two design approaches are radically different.
  - However, are complementary
    - rather than competing techniques.
  - Each technique is applicable at
    - different stages of the design process.

Function-Oriented Design

- A system is looked upon as something
  - that performs a set of functions.
- Starting at this high-level view of the system:
  - each function is successively refined into more detailed functions.
  - Functions are mapped to a module structure.

Example

- The function create-new-library-member:
  - creates the record for a new member,
  - assigns a unique membership number
  - prints a bill towards the membership

Example

- Create-library-member function consists of the following sub-functions:
  - assign-membership-number
  - create-member-record
  - print-bill

Function-Oriented Design

- Each subfunction:
  - split into more detailed subfunctions and so on.

Function-Oriented Design

- The system state is centralized:
  - accessible to different functions,
  - member-records:
    - available for reference and updation to several functions:
      - ✓ create-new-member
      - ✓ delete-member
      - ✓ update-member-record

Function-Oriented Design

- Several function-oriented design approaches have been developed:
  - Structured design (Constantine and Yourdon, 1979)
  - Jackson's structured design (Jackson, 1975)
  - Warnier-Orr methodology
  - Wirth's step-wise refinement
  - Hatley and Pirbhai's Methodology

Object-Oriented Design

- System is viewed as a collection of objects (i.e. entities) .
- System state is decentralized among the objects:
  - each object manages its own state information.

Object-Oriented Design Example

- Library Automation Software:
  - each library member is a separate object
    - with its own data and functions.
  - Functions defined for one object:
    - cannot directly refer to or change data of other objects.

Object-Oriented Design

- Objects have their own internal data:
  - defines their state.
- Similar objects constitute a class.
  - each object is a member of some class.
- Classes may inherit features
  - from a super class.
- Conceptually, objects communicate by message passing.

Object-Oriented versus Function-Oriented Design

- Unlike function-oriented design,
  - in OOD the basic abstraction is not functions such as "sort", "display", "track", etc.,
  - but real-world entities such as "employee", "picture", "machine", "radar system", etc.

Object-Oriented versus Function-Oriented Design

- In OOD:
  - software is not developed by designing functions such as:
    - ✓update-employee-record,
    - ✓get-employee-address, etc.
  - but by designing objects such as:
    - ✓employees,
    - ✓departments, etc.

Object-Oriented versus Function-Oriented Design

- Grady Booch sums up this fundamental difference saying:
  - ✓"Identify verbs if you are after procedural design and nouns if you are after object-oriented design."

Object-Oriented versus Function-Oriented Design

- In OOD:
  - state information is not shared in a centralized data.
  - but is distributed among the objects of the system.

Example:

- In an employee pay-roll system, the following can be global data:
  - names of the employees,
  - their code numbers,
  - basic salaries, etc.
- Whereas, in object oriented systems:
  - data is distributed among different employee objects of the system.

Object-Oriented versus Function-Oriented Design

- Objects communicate by message passing.
  - one object may discover the state information of another object by interrogating it.

Object-Oriented versus Function-Oriented Design

- Of course, somewhere or other the functions must be implemented:
  - the functions are usually associated with specific real-world entities (objects)
  - directly access only part of the system state information.

Object-Oriented versus Function-Oriented Design

- Function-oriented techniques group functions together if:
  - as a group, they constitute a higher level function.
- On the other hand, object-oriented techniques group functions together:
  - on the basis of the data they operate on.

Object-Oriented versus Function-Oriented Design

- To illustrate the differences between object-oriented and function-oriented design approaches,
  - let us consider an example ---
  - An automated fire-alarm system for a large building.

Fire-Alarm System:

- We need to develop a computerized fire alarm system for a large multi-storied building:
  - There are 80 floors and 1000 rooms in the building.

Fire-Alarm System:

- Different rooms of the building:
  - fitted with smoke detectors and fire alarms.
- The fire alarm system would monitor:
  - status of the smoke detectors.

Fire-Alarm System

- Whenever a fire condition is reported by any smoke detector:
  - the fire alarm system should:
    - ✓ determine the location from which the fire condition was reported
    - ✓ sound the alarms in the neighboring locations.

Fire-Alarm System

- The fire alarm system should:
  - flash an alarm message on the computer console:
    - fire fighting personnel man the console round the clock.

Fire-Alarm System

- After a fire condition has been successfully handled,
  - the fire alarm system should let fire fighting personnel reset the alarms.

Function-Oriented Approach:

- /* Global data (system state) accessible by various functions */

```
BOOL  detector_status[1000];
int   detector_locs[1000];
BOOL  alarm-status[1000]; /* alarm activated when status set */
int   alarm_locs[1000]; /* room number where alarm is located */
int   neighbor-alarms[1000][10];/*each detector has at most*/
                /* 10 neighboring alarm locations */
```

The functions which operate on the system state:

```
interrogate_detectors();
get_detector_location();
determine_neighbor();
ring_alarm();
reset_alarm();
report_fire_location();
```

Object-Oriented Approach:

- class detector
  - attributes: status, location, neighbors
  - operations: create, sense-status, get-location, find-neighbors
- class alarm
  - attributes: location, status
  - operations: create, ring-alarm, get_location, reset-alarm
- In the object-oriented program,
  - appropriate number of instances of the class detector and alarm should be created.

Object-Oriented versus Function-Oriented Design

- In the function-oriented program:
  - the system state is centralized
  - several functions accessing these data are defined.
- In the object oriented program,
  - the state information is distributed among various sensor and alarm objects.

Object-Oriented versus Function-Oriented Design

- Use OOD to design the classes:
  - then applies top-down function oriented techniques
    - to design the internal methods of classes.

Object-Oriented versus Function-Oriented Design

- Though outwardly a system may appear to have been developed in an object oriented fashion,
  - but inside each class there is a small hierarchy of functions designed in a top-down manner.

Summary

- We started with an overview of:
  - activities undertaken during the software design phase.
- We identified:
  - the information need to be produced at the end of the design phase:
    - so that the design can be easily implemented using a programming language.

Summary

- We characterized the features of a good software design by introducing the concepts of:
  - fan-in, fan-out,
  - cohesion, coupling,
  - abstraction, etc.

Summary

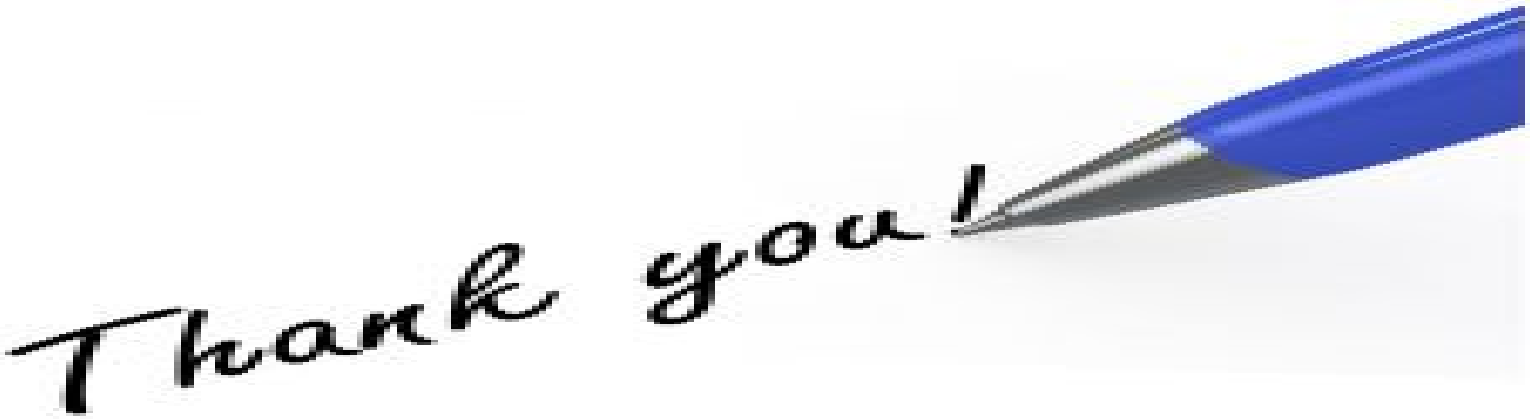
- We classified different types of cohesion and coupling:
  - enables us to approximately determine the cohesion and coupling existing in a design.

Summary

- Two fundamentally different approaches to software design:
  - ✓function-oriented approach
  - ✓object-oriented approach

Summary

- We looked at the essential philosophy behind these two approaches
  - these two approaches are not competing but complementary approaches.


Thank you!